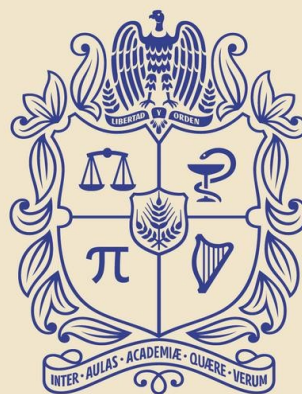




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OPTIMIZATION

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CIUDAD UNIVERSITARIA, BOGOTÁ D.C., APRIL 2026



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1. Introduction



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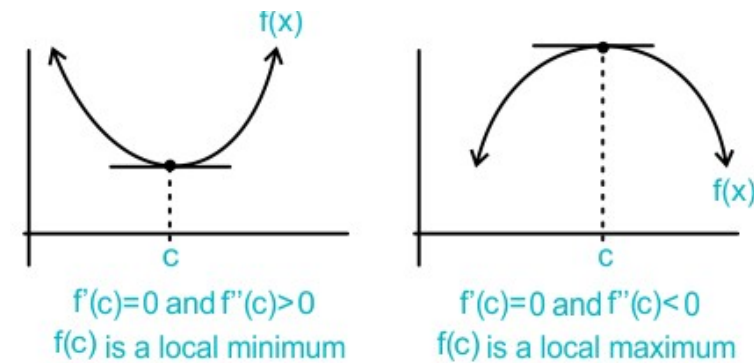


NUMERICAL METHODS



Relationship to Root Finding

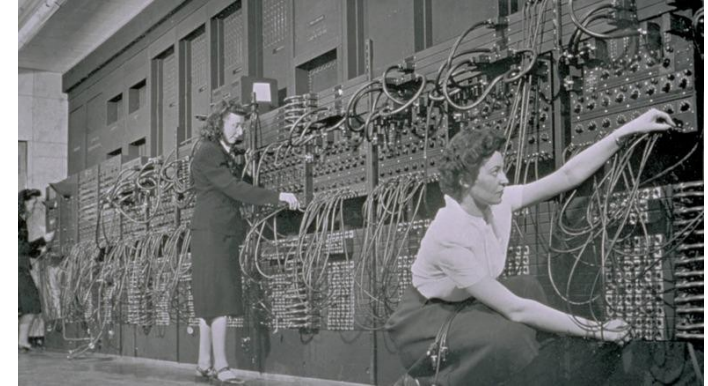
- Optimization seeks the points where a function's derivative $f'(x)$ equals zero and uses the second derivative $f''(x)$ to distinguish minima ($f''(x)>0$) from maxima ($f''(x)<0$).
- Many methods reduce optimization to solving the root problem $f'(x)=0$, often approximating derivatives numerically.





Historical and Non-Computer Methods

- Foundations laid by Bernoulli, Euler and Lagrange (calculus of variations and Lagrange multipliers for constrained problems).
- Post–World War II: Dantzig's simplex method for linear programming, followed by Charnes and colleagues, and rapid growth of unconstrained methods with computer availability.





Optimization in Engineering Practice

- Unlike descriptive (simulation) models, optimization models are **prescriptive**, prescribing the best design or course of action under physical and cost constraints.
- Engineers routinely balance performance, physical limits and cost.



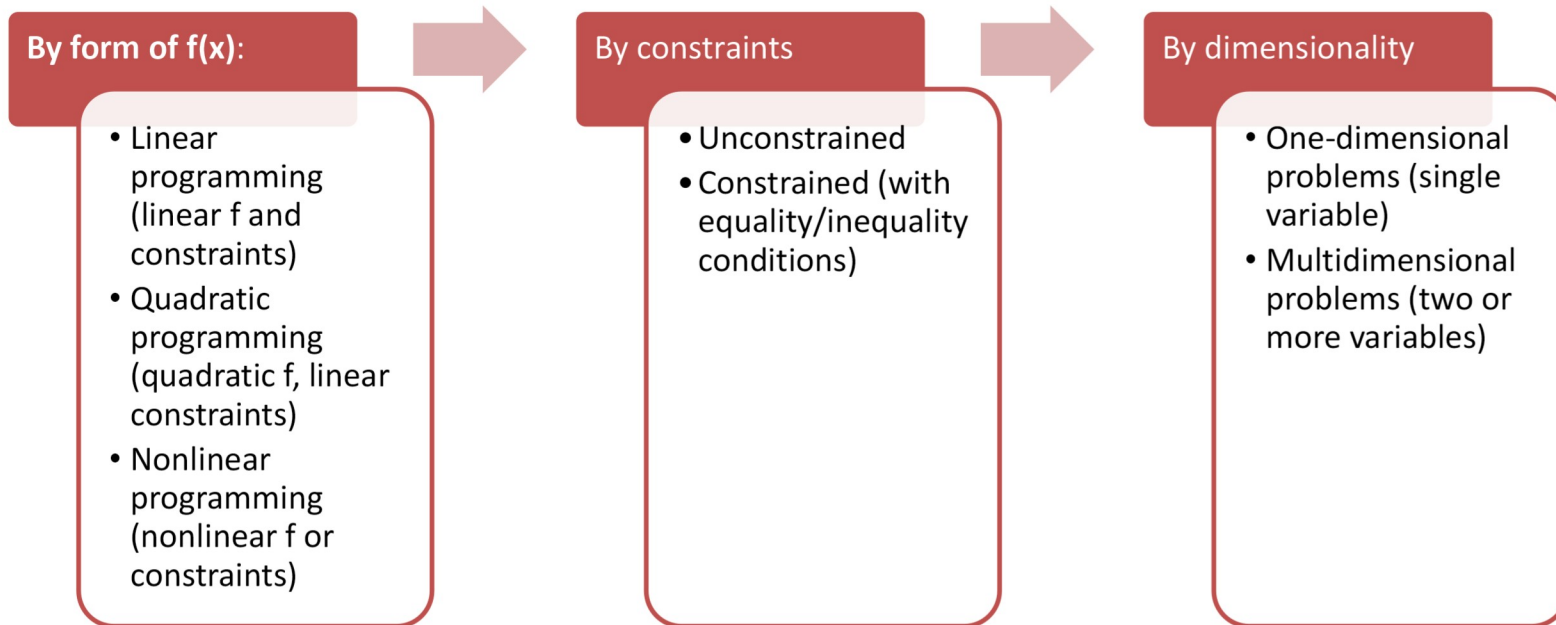


Common Engineering Optimization Examples

- Design aircraft for minimum weight and maximum strength.
 - Optimal trajectories of space vehicles.
 - Design civil engineering structures for minimum cost.
 - Design water-resource projects like dams to mitigate flood damage while yielding maximum hydropower.
 - Predict structural behavior by minimizing potential energy.
 - Material-cutting strategy for minimum cost.
 - Design pump and heat transfer equipment for maximum efficiency.
 - Maximize power output of electrical networks and machinery while minimizing heat generation.
 - Shortest route of salesperson visiting various cities during one sales trip.
 - Optimal planning and scheduling.
 - Statistical analysis and models with minimum error.
 - Optimal pipeline networks.
 - Inventory control.
 - Maintenance planning to minimize cost.
 - Minimize waiting and idling times.
 - Design waste treatment systems to meet water-quality standards at least cost.
-



Classification of Optimization Problems





2. One-Dimensional Unconstrained Optimization



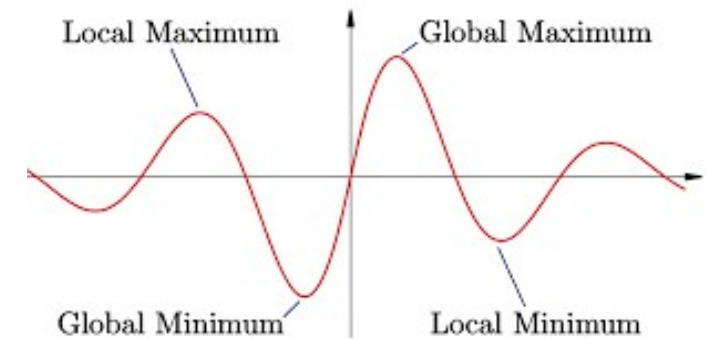
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Overview of One-Dimensional Unconstrained Optimization

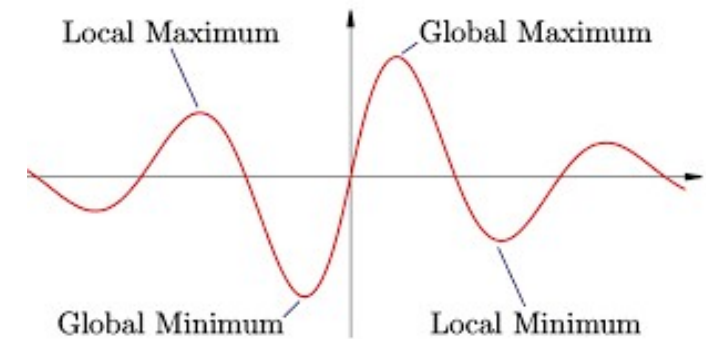
- Goal: find the minimum or maximum of a single-variable function $f(x)$.
- Challenges: functions can be **multimodal**, with multiple local extrema; distinguishing the **global** extremum is nontrivial.





General Strategies for Locating a Global Extremum

1. **Graphical Insight** – visualize low-dimensional functions to identify likely global optima.
2. **Multiple Starting Guesses** – run a local optimizer from varied (possibly random) starting points and select the best result.
3. **Perturbation** – after converging to a local extremum, slightly perturb the solution and rerun to check for better optima.





Classification of One-Dimensional Optimization Methods

- **Bracketing Methods** (require an initial interval $[x_l, x_u]$ containing a single extremum):
 - *Golden-Section Search*
 - *Parabolic Interpolation* (faster convergence but risk of divergence)
- **Open (Point) Methods** (rely on derivatives):
 - *Newton's Method* (solve $f'(x)=0$ as a root-finding problem)
- **Hybrid Methods**:
 - *Brent's Method* (combines golden-section reliability with parabolic speed)



Golden-Section Search Key Concepts

1. **Unimodal Interval** – assume $[x_l, x_u]$

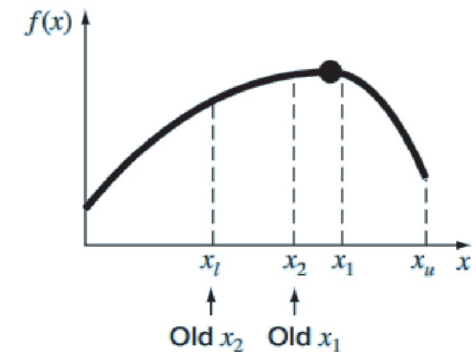
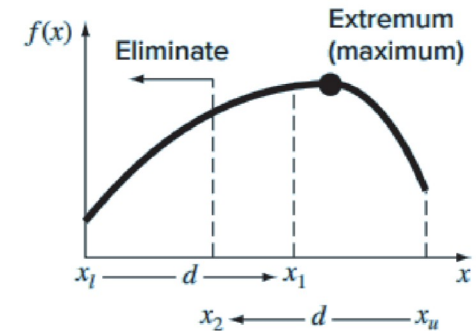
2. **Interior Points** – choose

$$d = \frac{\sqrt{5} - 1}{2} (x_u - x_l), \quad x_1 = x_l + d, \quad x_2 = x_u - d$$

3. **Interval Reduction** – evaluate $f(x_1)$ and $f(x_2)$:

- If $f(x_1) > f(x_2)$ (seeking maximum), discard $[x_l, x_2]$.
- Otherwise, discard $[x_1, x_u]$.

4. **Function-Evaluation Savings** – by reusing the interior point from the previous iteration, only one new evaluation per step is needed.





Example 1

Use the golden-section search to find the maximum of

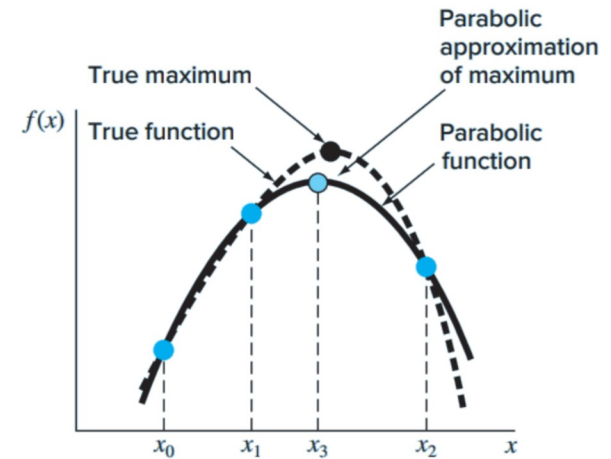
$$f(x) = 2 \sin x - \frac{x^2}{10}$$

within the interval $x_l = 0$ and $x_u = 4$



Parabolic Interpolation

- **Purpose:** Approximate the shape of a function $f(x)$ near its extremum using a quadratic (parabola).
- **Key idea:** Just as a straight line is uniquely defined by two points, a parabola is uniquely defined by three points.
- Formula:
$$x_3 = \frac{f(x_0)(x_1^2 - x_2^2) + f(x_1)(x_2^2 - x_0^2) + f(x_2)(x_0^2 - x_1^2)}{2[f(x_0)(x_1 - x_2) + f(x_1)(x_2 - x_0) + f(x_2)(x_0 - x_1)]}$$
- Where x_0, x_1, x_2 are three guesses bracketing the optimum, and x_3 is the estimated extremum.





Newton's Method for Optimization

- **Background:** Adaptation of Newton–Raphson root-finding to extrema of $f(x)$.
- **Iteration formula:**

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

- **Characteristics:**
 - **Open method:** Does not require initial bracketing of the optimum.
 - **Rapid (quadratic) convergence** when close to the solution.
 - **Potential divergence** if the initial guess is poor or if $f''(x_i)$ is zero/changes sign.
- **Practical advice:** Always check that the second derivative has the desired sign (positive for a minimum, negative for a maximum) to ensure convergence to the intended extremum.



Example 2

Use the parabolic interpolation and Newton's Method to find the maximum of

$$f(x) = 2 \sin x - \frac{x^2}{10}$$

For parabolic interpolation use an initial guesses of $x_0 = 0$, $x_1 = 1$, and $x_2 = 4$. For Newton's Method use an initial guess of $x_0 = 2.5$



3. Multidimensional Unconstrained Optimization



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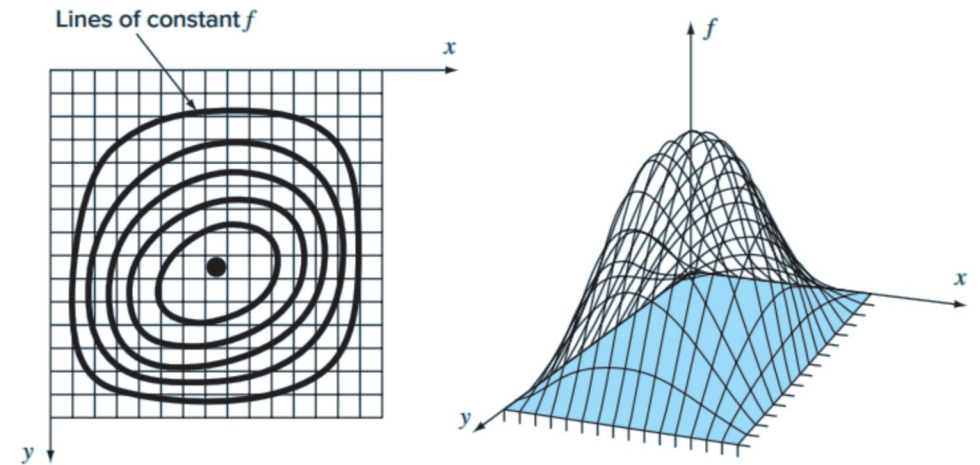
NUMERICAL METHODS

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Overview: Multidimensional Unconstrained Optimization

- Goal: Find minima or maxima of functions of several variables.
- Focus here: Two-dimensional case (mountains and valleys analogy).
- Classification of methods:
 - **Non-gradient (Direct) methods**
 - **Gradient (Descent/Ascent) methods**





Direct Methods (Non-gradient)

Random Search

- Brute-force sampling of function at random points.
- Guarantees finding the global optimum eventually.
- Drawbacks:
 - Highly inefficient as dimensionality grows.
 - Ignores any information about function behavior.

More Sophisticated Heuristics

- Simulated annealing
- Tabu search
- Artificial neural networks
- **Genetic algorithms** (most widely used; pioneered by Holland 1975; overviews by Davis 1991 and Goldberg 1989)



Example 3

Use a random number generator to locate the maximum of

$$f(x, y) = y - x - 2x^2 - 2xy - y^2$$

in the domain bounded by $x = -2$ to 2 and $y = 1$ to 3 .



4. Constrained Optimization



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Constrained Optimization Overview

- Deals with optimization problems where constraints play a role.
- When both the objective function and the constraints are linear, specialized methods (linear programming) can solve very large problems efficiently.
- Applicable in diverse engineering and management settings.
- Nonlinear constrained optimization is discussed more briefly, and software package options are surveyed.



Linear Programming (LP) Definition

- An LP problem seeks to maximize (or minimize) a linear objective (e.g., profit or cost) subject to linear constraints representing limited resources.
- “Linear” refers to the fact that both the objective and all constraints are linear functions of the decision variables.
- “Programming” here means “scheduling” or “setting an agenda,” not computer programming.



Graphical Solution (for Two-Variable LPs)

- Plot each linear constraint as a straight line in the $x_1 - x_2$ plane (with x_1 on the horizontal axis, x_2 on the vertical axis).
- The region where all constraints overlap (including $x_1 \geq 0$ and $x_2 \geq 0$) is the feasible solution space, containing all points that satisfy every constraint.



Example 4

Develop a graphical solution for the following problem:

$$\text{Maximize } Z = 150x_1 + 175x_2$$

subject to

$$7x_1 + 11x_2 \leq 77$$

$$10x_1 + 8x_2 \leq 80$$

$$x_1 \leq 9$$

$$x_2 \leq 6$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$



5. Case Studies: Optimization



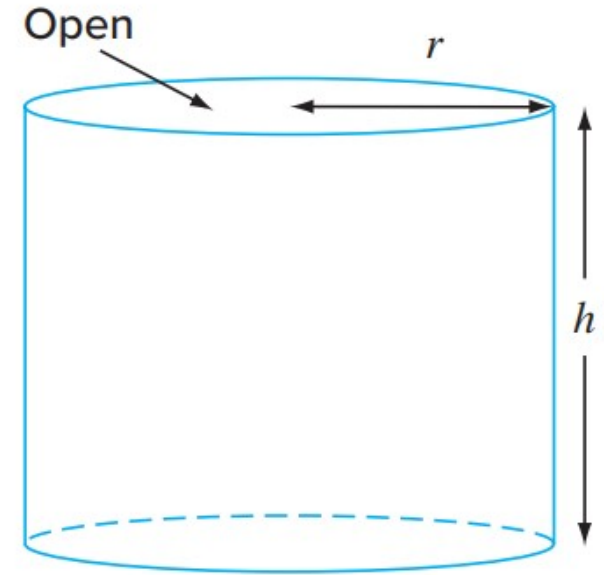
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Example 5

Design the optimal cylindrical container that is open at one end and has walls of negligible thickness. The container is to hold 0.2 m^3 . Design it so that the areas of its bottom and sides are minimized.





Example 6

A finite-element model of a cantilever beam subject to loading and moments is given by optimizing

$$f(x, y) = 5x^2 - 5xy + 2.5y^2 - x - 1.5y$$

where x = end displacement and y = end moment. Find the values of x and y that minimize $f(x, y)$.



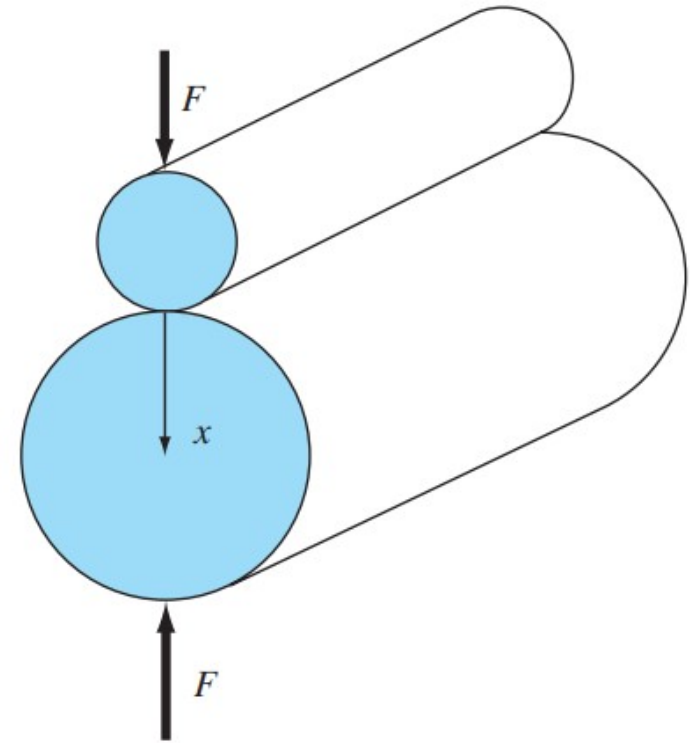


Example 7

Roller bearings are subject to fatigue failure caused by large contact loads F . The problem of finding the location of the maximum stress along the x axis can be shown to be equivalent to maximizing the function

$$f(x) = \frac{0.4}{\sqrt{1+x^2}} - \sqrt{1+x^2} \left(1 - \frac{0.4}{1+x^2} \right) + x$$

Find the value of x that maximizes $f(x)$ with Golden-Section Search.





6. Summary



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Summary

- **Definition of Optimization**

Finding points where a function's derivative $f'(x)=0$ and using $f''(x)$ to distinguish minima ($f''>0$) from maxima ($f''<0$); many algorithms reduce to solving root-finding problems and approximate derivatives numerically .

- **Historical Foundations**

Early work by Bernoulli, Euler, and Lagrange (calculus of variations, Lagrange multipliers); post–WWII advances include Dantzig's simplex method for linear programming and rapid growth of unconstrained methods with the advent of computers .

- **Role in Engineering Practice**

Optimization models are prescriptive (unlike descriptive simulation), prescribing best designs or actions under physical and cost constraints; engineers balance performance, physical limits, and cost .



Summary

- **One-Dimensional Unconstrained Optimization**
 - **Goal:** Find global minima/maxima of single-variable functions, which may be multimodal.
 - **Methods:**
 - Bracketing:* Golden-section search; parabolic interpolation
 - Open (derivative-based):* Newton's method
- **Multidimensional Unconstrained Optimization**
 - Direct (non-gradient) methods:** Random search
- **Constrained Optimization**
 - **Linear Programming (LP):** Maximize/minimize a linear objective subject to linear constraints (“programming” meaning scheduling)
 - **Solution approaches:** Simplex method; graphical solutions for two variables